Pi Panel
Restoration & troubleshooting handbook

Twelve operating modes. Six switches. Four sockets. A systematic approach to identifying the selected configuration, restoring its inputs, and confirming completion.

Purpose of this handbook

Use this handbook at the physical demonstration panel. The Power App/Page selects the scenario; the panel controls supply the input states. Each scenario assigns its own meaning and required state to those controls.

Begin with the contents and operating method. Then use the chapter matching the selected mode. A procedure for another mode can leave apparently reasonable settings on the panel without satisfying the current restoration condition.

How the information is organized

Each mode has seven sections: operating context, control interpretation, symptom assessment, restoration steps 1-5, restoration steps 6-10, acceptance checks, and unresolved-fault review. Cross-references identify the next page to consult.

Scope

This handbook covers the Pi demonstration panel. Device names and functional labels describe game scenarios; the procedures operate the demonstration switches and sockets.

Print and binding

100 pages. US Letter, portrait, actual size (100%). Use Fit for A4. Print single-sided for a 100-sheet participant binder, or double-sided for 50 sheets. Black-and-white printing is supported.

Document reference: PP-RH-100 | Expanded participant edition

READER NAVIGATION

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Each mode entry points to its chapter map. Control indexes locate individual steps.

Operating method

1. Establish the selected mode

Read the selection in the Power App/Page and find the matching chapter. Identify both its number and name. Do not infer the mode from positions left by a previous participant: a position can be correct for one mode and incorrect for another.

2. Interpret the labels

Locate the control label on the panel, then consult the chapter dictionary. SW1-SW6 identify switches; J1-J4 identify socket inputs. Functional wording describes the scenario. Physical labels remain the same when the scenario changes.

3. Record the initial condition

Observe before adjusting. Record six switch positions, four socket conditions, and four indicator states. Mark uncertain readings unknown. A written observation is more useful than an assumption about the previous participant.

4. Work through restoration

Read steps 1-5 and then 6-10. Apply the action at the named control and verify its position. If it already matches, confirm it and continue. Keep earlier corrected controls in their required states.

5. Verify the complete configuration

LED4 green is the completion indication. The documented order provides a repeatable method; the Pi checks the current combination of input states. It does not use printed step numbers as a historical sequence lock.

6. Resolve exceptions

If completion is absent, use the chapter acceptance and unresolved-fault sections. If panel and screen disagree, record both observations and ask the facilitator to investigate the setup.

Switch identification

Label	Physical control	State language
SW1	Safety switch 1	ON / closed or OFF / open
SW2	Safety switch 2	ON / closed or OFF / open
SW3	Toggle switch 1	ON / closed or OFF / open
SW4	Toggle switch 2	ON / closed or OFF / open
SW5	Keyed switch 1	ON / closed or OFF / open
SW6	Keyed switch 2	ON / closed or OFF / open

Physical identity and scenario function

Start with the SW number. A description such as power, service key, alarm arm, or cloud sync belongs to a particular mode. It gives a scenario meaning, not an additional control. The chapter dictionary joins physical identity to that meaning.

Reading ON and OFF

ON means the closed input state. OFF means the open input state. Do not assume ON is always the target. Some modes require an override, inspection setting, lockout, or another input to be open. Follow the explicit instruction.

Checking a switch

Locate the label, read the desired state, operate that control if necessary, and confirm its position. If markings do not let you determine ON and OFF, ask the facilitator to identify the positions. Do not substitute a similar-looking switch.

Switch index: page 93.

Socket interpretation

Four named inputs

The socket inputs are J1, J2, J3, and J4. A procedure can require some connected and others open. A connected socket is not automatically correct; the selected mode determines the required combination.

Instruction	Required condition	Confirmation
Plug in	Connected	Confirm the named socket is connected.
Unplug	Open	Confirm the named socket is left open.
Already correct	Keep the condition	Verify its identifier before continuing.

What LED3 tells you

Orange reports at least one connected socket input. It does not identify the socket, count connections, or certify the correct combination. An extra connected socket can keep orange on while preventing restoration.

After changing a cable

Inspect every input involved in the change. A cable arrangement may involve more than one connection; the procedure still specifies the state of each named input. Recheck earlier steps if a later adjustment disturbed them.

Cable preparation is separate

The appendix preserves the demo bundle pinout reference. Those pinouts are independent of the Pi harness. During restoration, work from socket labels and required states in the active chapter.

Socket index: page 95. Cable reference: page 97.

Indicator interpretation

Light	Meaning when lit	Does not establish
LED1 / Blue	Ready/power condition active.	All ten inputs match.
LED2 / Yellow	Restoration still in progress.	Which input is incorrect.
LED3 / Orange	At least one socket connected.	Correct socket combination.
LED4 / Green	Current mode restored.	Another mode is restored.

Interpret lights with the mode

Confirm the selection before judging a result. A successful configuration belongs to that mode. Changing the mode can change whether existing positions satisfy restoration without physically moving a control.

Blue with yellow

Readiness can be present while other conditions are missing. Continue the full procedure. Do not skip keyed switches, toggles, or sockets just because the ready/power light is present.

Orange with yellow

At least one socket input is connected, but restoration is incomplete. Check socket instructions, including required-open sockets, then the remaining switch instructions.

Green absent after all steps

Compare all ten conditions with the procedure. If they appear correct, preserve observations and use unresolved-fault review. A light alone cannot identify whether the cause is a mistaken position, a detected input, or the setup.

Initial condition survey

Establish what is selected and what is physically present so later observations can be compared with a recorded starting point.

Selection and chapter

Write the mode name. Locate its chapter and check the title against the screen. Leave controls unchanged until the selection is clear. If the selection changes during a run, re-establish the mode before continuing.

Physical controls

Read SW1-SW6 in numerical order and record ON, OFF, or unknown. Then read J1-J4 and record connected, open, or unknown. This survey order records observations; the restoration order is in the chapter.

Panel indicators

Record whether LED1-LED4 are lit. Use each identifier as well as its color. If green is already present, tell the facilitator before starting a timed restoration: the configuration may already be restored.

Observation versus interpretation

Write "J2 connected" as an observation. Write "J2 may be incorrect for this mode" as an interpretation until the procedure confirms its target. A suspected cause is not a measured condition.

Recording uncertainty

If you cannot read a label, determine a position, or confirm a mode, record the uncertainty. Random adjustments make it harder to reconstruct the state. Resolve the unknown or involve the facilitator.

Worksheet: page 99. Diagnostic routing: page 8.

Diagnostic routing

Observation	Next action
Selected mode unclear	Confirm selection before using a chapter.
Green at the start	Ask the facilitator to confirm the initial condition.
Mode known; incomplete	Read the chapter symptom assessment.
Blue lit; green absent	Continue the procedure; readiness is not completion.
Orange lit; green absent	Check required-open and required-connected sockets.
All targets appear correct	Use acceptance and unresolved-fault checks.
Panel and screen disagree	Record both; refer the setup to the facilitator.

Several symptoms at once

Use the mode chapter as the organizing reference. An indicator can narrow the next inspection but cannot replace the ten-input procedure. Avoid changing chapters because another label seems related to the symptom.

An adjustment has no obvious effect

An adjustment need not produce a unique lamp change. Confirm the physical state and continue. LED4 checks the combined result; intermediate lights are not a per-step progress counter.

Completion is lost

If green goes out, check whether a control moved or the mode changed. Reconfirm the current configuration. A written record describes a past observation, not a permanent lock on the panel.

Smart Thermostat

This chapter applies when **Smart Thermostat** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Thermostat restored - holding your set temperature.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 3 switches closed and 3 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: thermostat

Control interpretation

The table maps physical labels to meanings for Smart Thermostat. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - eco lockout
SW3	Toggle switch 1 - fan running
SW4	Toggle switch 2 - heat/cool select
SW5	Keyed switch 1
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 11. Restoration starts on page 12.

Symptom assessment

Confirm **Smart Thermostat** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 5 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 12. If every step is complete, use acceptance checks on page 14. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Smart Thermostat**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - fan running. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 2 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 3 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J4 / Unplug

Locate J4, described as socket j4 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW5 / Set ON

Locate SW5, identified as keyed switch 1. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 13.

Restoration procedure: 6-10

Apply these steps to **Smart Thermostat**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 7 / SW2 / Set OFF

Locate SW2, identified as safety switch 2 - eco lockout. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 8 / SW6 / Set OFF

Locate SW6, identified as keyed switch 2. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / J3 / Unplug

Locate J3, described as socket j3 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 10 / SW4 / Set OFF

Locate SW4, identified as toggle switch 2 - heat/cool select. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 14. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Smart Thermostat**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 3 switches closed and 3 open. Review each SW instruction on pages 12 and 13. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Thermostat restored - holding your set temperature.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 15. Restoration record: page 100.

Unresolved-fault review

Use this review when the Smart Thermostat procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW2, SW6, SW4, J4, J3**. Verify their actions on pages 12 and 13. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW3, SW1, SW5, J1, J2**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 14 after checking any new adjustment.

Smart Lock & Security Panel

This chapter applies when **Smart Lock & Security Panel** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

House is secured now.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 5 switches closed and 1 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: smart-lock

Control interpretation

The table maps physical labels to meanings for Smart Lock & Security Panel. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - deadbolt
SW3	Toggle switch 1 - auto-lock
SW4	Toggle switch 2 - alarm arm
SW5	Keyed switch 1 - service key
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 18. Restoration starts on page 19.

Symptom assessment

Confirm **Smart Lock & Security Panel** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 19. If every step is complete, use acceptance checks on page 21. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Smart Lock & Security Panel**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / J2 / Unplug

Locate J2, described as socket j2 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 2 / SW5 / Set ON

Locate SW5, identified as keyed switch 1 - service key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - deadbolt. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J4 / Plug in

Locate J4, described as socket j4 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 20.

Restoration procedure: 6-10

Apply these steps to **Smart Lock & Security Panel**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - alarm arm. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J1 / Unplug

Locate J1, described as socket j1 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW3 / Set OFF

Locate SW3, identified as toggle switch 1 - auto-lock. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / SW6 / Set ON

Locate SW6, identified as keyed switch 2. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 10 / J3 / Plug in

Locate J3, described as socket j3 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Perform acceptance checks on page 21. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Smart Lock & Security Panel**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 19 and 20. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

House is secured now.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 22. Restoration record: page 100.

Unresolved-fault review

Use this review when the Smart Lock & Security Panel procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW3, J2, J1**. Verify their actions on pages 19 and 20. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW5, SW2, SW1, SW4, SW6, J4, J3**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 21 after checking any new adjustment.

Cold-Chain Warehouse Sensor

This chapter applies when **Cold-Chain Warehouse Sensor** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Cold storage is stable and monitoring is back online.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 3 switches closed and 3 open; 3 socket inputs connected and 1 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: coldchain-sensor

Control interpretation

The table maps physical labels to meanings for Cold-Chain Warehouse Sensor. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - calibration
SW3	Toggle switch 1 - threshold
SW4	Toggle switch 2 - cloud sync
SW5	Keyed switch 1
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 25. Restoration starts on page 26.

Symptom assessment

Confirm **Cold-Chain Warehouse Sensor** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 3 connected and 1 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 4 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 26. If every step is complete, use acceptance checks on page 28. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Cold-Chain Warehouse Sensor**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - cloud sync. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 2 / J3 / Plug in

Locate J3, described as socket j3 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 3 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / SW6 / Set OFF

Locate SW6, identified as keyed switch 2. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 5 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Continue on page 27.

Restoration procedure: 6-10

Apply these steps to **Cold-Chain Warehouse Sensor**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - calibration. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J4 / Unplug

Locate J4, described as socket j4 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW5 / Set OFF

Locate SW5, identified as keyed switch 1. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / SW3 / Set OFF

Locate SW3, identified as toggle switch 1 - threshold. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 10 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Perform acceptance checks on page 28. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Cold-Chain Warehouse Sensor**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 3 switches closed and 3 open. Review each SW instruction on pages 26 and 27. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 3 inputs connected and 1 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Cold storage is stable and monitoring is back online.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 29. Restoration record: page 100.

Unresolved-fault review

Use this review when the Cold-Chain Warehouse Sensor procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW6, SW5, SW3, J4**. Verify their actions on pages 26 and 27. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW4, SW1, SW2, J3, J1, J2**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 28 after checking any new adjustment.

Water Pump Station Controller

This chapter applies when **Water Pump Station Controller** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Pump station is back online and operating normally.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 4 switches closed and 2 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: water-pump

Control interpretation

The table maps physical labels to meanings for Water Pump Station Controller. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - pump enable
SW3	Toggle switch 1 - pressure relief
SW4	Toggle switch 2 - remote override
SW5	Keyed switch 1
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 32. Restoration starts on page 33.

Symptom assessment

Confirm **Water Pump Station Controller** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 4 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 33. If every step is complete, use acceptance checks on page 35. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Water Pump Station Controller**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / J4 / Plug in

Locate J4, described as socket j4 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 2 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - pump enable. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / SW5 / Set OFF

Locate SW5, identified as keyed switch 1. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J1 / Unplug

Locate J1, described as socket j1 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW4 / Set OFF

Locate SW4, identified as toggle switch 2 - remote override. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 34.

Restoration procedure: 6-10

Apply these steps to **Water Pump Station Controller**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 7 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 8 / J3 / Unplug

Locate J3, described as socket j3 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 9 / SW6 / Set ON

Locate SW6, identified as keyed switch 2. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 10 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - pressure relief. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 35. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Water Pump Station Controller**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 4 switches closed and 2 open. Review each SW instruction on pages 33 and 34. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Pump station is back online and operating normally.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 36. Restoration record: page 100.

Unresolved-fault review

Use this review when the Water Pump Station Controller procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW5, SW4, J1, J3**. Verify their actions on pages 33 and 34. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW2, SW1, SW6, SW3, J4, J2**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 35 after checking any new adjustment.

Tactical Field Radio

This chapter applies when **Tactical Field Radio** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Secure link established with base.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 5 switches closed and 1 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: field-radio

Control interpretation

The table maps physical labels to meanings for Tactical Field Radio. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - encryption
SW3	Toggle switch 1 - channel hop
SW4	Toggle switch 2 - transmit
SW5	Keyed switch 1
SW6	Keyed switch 2 - crypto key
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 39. Restoration starts on page 40.

Symptom assessment

Confirm **Tactical Field Radio** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 40. If every step is complete, use acceptance checks on page 42. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Tactical Field Radio**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / SW6 / Set ON

Locate SW6, identified as keyed switch 2 - crypto key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 2 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 3 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - channel hop. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J1 / Unplug

Locate J1, described as socket j1 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 41.

Restoration procedure: 6-10

Apply these steps to **Tactical Field Radio**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / J4 / Unplug

Locate J4, described as socket j4 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 7 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - encryption. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 8 / SW5 / Set OFF

Locate SW5, identified as keyed switch 1. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / J3 / Plug in

Locate J3, described as socket j3 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 10 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - transmit. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 42. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Tactical Field Radio**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 40 and 41. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Secure link established with base.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 43. Restoration record: page 100.

Unresolved-fault review

Use this review when the Tactical Field Radio procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW5, J1, J4**. Verify their actions on pages 40 and 41. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW6, SW3, SW1, SW2, SW4, J2, J3**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 42 after checking any new adjustment.

Smart EV Charger

This chapter applies when **Smart EV Charger** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

EV charger restored - charging resumed.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 5 switches closed and 1 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: ev-charger

Control interpretation

The table maps physical labels to meanings for Smart EV Charger. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - contactor
SW3	Toggle switch 1 - ground check
SW4	Toggle switch 2 - billing link
SW5	Keyed switch 1 - service lock
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 46. Restoration starts on page 47.

Symptom assessment

Confirm **Smart EV Charger** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 47. If every step is complete, use acceptance checks on page 49. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Smart EV Charger**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / J3 / Unplug

Locate J3, described as socket j3 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 2 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / SW5 / Set ON

Locate SW5, identified as keyed switch 1 - service lock. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - billing link. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 5 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Continue on page 48.

Restoration procedure: 6-10

Apply these steps to **Smart EV Charger**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - contactor. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J4 / Plug in

Locate J4, described as socket j4 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW6 / Set OFF

Locate SW6, identified as keyed switch 2. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - ground check. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 10 / J2 / Unplug

Locate J2, described as socket j2 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Perform acceptance checks on page 49. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Smart EV Charger**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 47 and 48. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

EV charger restored - charging resumed.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 50. Restoration record: page 100.

Unresolved-fault review

Use this review when the Smart EV Charger procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW6, J3, J2**. Verify their actions on pages 47 and 48. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW1, SW5, SW4, SW2, SW3, J1, J4**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 49 after checking any new adjustment.

Elevator Control Panel

This chapter applies when **Elevator Control Panel** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Elevator back in service.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 4 switches closed and 2 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: elevator

Control interpretation

The table maps physical labels to meanings for Elevator Control Panel. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - brake release
SW3	Toggle switch 1 - doors closed
SW4	Toggle switch 2 - inspection mode
SW5	Keyed switch 1
SW6	Keyed switch 2 - inspection key
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 53. Restoration starts on page 54.

Symptom assessment

Confirm **Elevator Control Panel** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 4 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 54. If every step is complete, use acceptance checks on page 56. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Elevator Control Panel**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - brake release. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 2 / J4 / Unplug

Locate J4, described as socket j4 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 3 / SW6 / Set ON

Locate SW6, identified as keyed switch 2 - inspection key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 55.

Restoration procedure: 6-10

Apply these steps to **Elevator Control Panel**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - doors closed. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW5 / Set OFF

Locate SW5, identified as keyed switch 1. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / J3 / Unplug

Locate J3, described as socket j3 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 10 / SW4 / Set OFF

Locate SW4, identified as toggle switch 2 - inspection mode. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 56. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Elevator Control Panel**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 4 switches closed and 2 open. Review each SW instruction on pages 54 and 55. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Elevator back in service.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 57. Restoration record: page 100.

Unresolved-fault review

Use this review when the Elevator Control Panel procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW5, SW4, J4, J3**. Verify their actions on pages 54 and 55. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW2, SW6, SW1, SW3, J2, J1**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 56 after checking any new adjustment.

Traffic Signal Controller

This chapter applies when **Traffic Signal Controller** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Traffic signal restored to normal cycle.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 4 switches closed and 2 open; 3 socket inputs connected and 1 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: traffic-signal

Control interpretation

The table maps physical labels to meanings for Traffic Signal Controller. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - controller
SW3	Toggle switch 1 - timing plan
SW4	Toggle switch 2 - manual override
SW5	Keyed switch 1 - cabinet key
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 60. Restoration starts on page 61.

Symptom assessment

Confirm **Traffic Signal Controller** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 3 connected and 1 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 61. If every step is complete, use acceptance checks on page 63. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Traffic Signal Controller**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 2 / SW4 / Set OFF

Locate SW4, identified as toggle switch 2 - manual override. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J3 / Plug in

Locate J3, described as socket j3 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW5 / Set ON

Locate SW5, identified as keyed switch 1 - cabinet key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 62.

Restoration procedure: 6-10

Apply these steps to **Traffic Signal Controller**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - controller. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J2 / Unplug

Locate J2, described as socket j2 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW6 / Set OFF

Locate SW6, identified as keyed switch 2. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - timing plan. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 10 / J4 / Plug in

Locate J4, described as socket j4 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Perform acceptance checks on page 63. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Traffic Signal Controller**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 4 switches closed and 2 open. Review each SW instruction on pages 61 and 62. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 3 inputs connected and 1 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Traffic signal restored to normal cycle.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 64. Restoration record: page 100.

Unresolved-fault review

Use this review when the Traffic Signal Controller procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW4, SW6, J2**. Verify their actions on pages 61 and 62. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW1, SW5, SW2, SW3, J1, J3, J4**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 63 after checking any new adjustment.

Hospital Patient Monitor

This chapter applies when **Hospital Patient Monitor** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Patient monitor restored - vitals streaming normally.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 5 switches closed and 1 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: patient-monitor

Control interpretation

The table maps physical labels to meanings for Hospital Patient Monitor. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - leads connected
SW3	Toggle switch 1 - SpO2 sensor
SW4	Toggle switch 2 - alarm audible
SW5	Keyed switch 1 - nurse key
SW6	Keyed switch 2
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 67. Restoration starts on page 68.

Symptom assessment

Confirm **Hospital Patient Monitor** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 68. If every step is complete, use acceptance checks on page 70. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Hospital Patient Monitor**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / SW5 / Set ON

Locate SW5, identified as keyed switch 1 - nurse key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 2 / J3 / Unplug

Locate J3, described as socket j3 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 3 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - leads connected. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - alarm audible. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 69.

Restoration procedure: 6-10

Apply these steps to **Hospital Patient Monitor**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J4 / Unplug

Locate J4, described as socket j4 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - spo2 sensor. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / SW6 / Set OFF

Locate SW6, identified as keyed switch 2. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 10 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Perform acceptance checks on page 70. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Hospital Patient Monitor**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 68 and 69. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Patient monitor restored - vitals streaming normally.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 71. Restoration record: page 100.

Unresolved-fault review

Use this review when the Hospital Patient Monitor procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW6, J3, J4**. Verify their actions on pages 68 and 69. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW5, SW2, SW4, SW1, SW3, J1, J2**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 70 after checking any new adjustment.

Retail Payment Kiosk

This chapter applies when **Retail Payment Kiosk** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Payment kiosk back online and processing.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 5 switches closed and 1 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: retail-kiosk

Control interpretation

The table maps physical labels to meanings for Retail Payment Kiosk. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - receipt printer
SW3	Toggle switch 1 - card reader
SW4	Toggle switch 2 - network
SW5	Keyed switch 1
SW6	Keyed switch 2 - cash drawer key
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 74. Restoration starts on page 75.

Symptom assessment

Confirm **Retail Payment Kiosk** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 75. If every step is complete, use acceptance checks on page 77. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Retail Payment Kiosk**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - card reader. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 2 / SW6 / Set ON

Locate SW6, identified as keyed switch 2 - cash drawer key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / J4 / Plug in

Locate J4, described as socket j4 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 4 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 5 / J2 / Unplug

Locate J2, described as socket j2 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Continue on page 76.

Restoration procedure: 6-10

Apply these steps to **Retail Payment Kiosk**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - network. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 7 / J1 / Unplug

Locate J1, described as socket j1 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 8 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - receipt printer. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / J3 / Plug in

Locate J3, described as socket j3 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 10 / SW5 / Set OFF

Locate SW5, identified as keyed switch 1. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 77. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Retail Payment Kiosk**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 75 and 76. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Payment kiosk back online and processing.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 78. Restoration record: page 100.

Unresolved-fault review

Use this review when the Retail Payment Kiosk procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW5, J2, J1**. Verify their actions on pages 75 and 76. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW3, SW6, SW1, SW4, SW2, J4, J3**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 77 after checking any new adjustment.

Drone Ground Control

This chapter applies when **Drone Ground Control** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Drone link restored - ready for flight ops.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

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Configuration characteristics

This mode requires 5 switches closed and 1 open; 3 socket inputs connected and 1 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: drone-ground-control

Control interpretation

The table maps physical labels to meanings for Drone Ground Control. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - power
SW2	Safety switch 2 - GPS lock
SW3	Toggle switch 1 - telemetry
SW4	Toggle switch 2 - uplink
SW5	Keyed switch 1
SW6	Keyed switch 2 - flight key
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 81. Restoration starts on page 82.

Symptom assessment

Confirm **Drone Ground Control** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 3 connected and 1 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 2 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 82. If every step is complete, use acceptance checks on page 84. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Drone Ground Control**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 2 / SW4 / Set ON

Locate SW4, identified as toggle switch 2 - uplink. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / SW6 / Set ON

Locate SW6, identified as keyed switch 2 - flight key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J1 / Plug in

Locate J1, described as socket j1 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - gps lock. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 83.

Restoration procedure: 6-10

Apply these steps to **Drone Ground Control**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / J3 / Unplug

Locate J3, described as socket j3 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 7 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 8 / SW5 / Set OFF

Locate SW5, identified as keyed switch 1. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / J4 / Plug in

Locate J4, described as socket j4 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 10 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - telemetry. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 84. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Drone Ground Control**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 82 and 83. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 3 inputs connected and 1 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Drone link restored - ready for flight ops.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 85. Restoration record: page 100.

Unresolved-fault review

Use this review when the Drone Ground Control procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW5, J3**. Verify their actions on pages 82 and 83. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW4, SW6, SW2, SW1, SW3, J2, J1, J4**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 84 after checking any new adjustment.

Industrial Robot Arm

This chapter applies when **Industrial Robot Arm** is selected in the Power App/Page. The panel has six switches and four socket inputs. This mode assigns their scenario meanings and required states.

Restoration objective

Robot cell restored - supervised automation is ready.

Use LED4 green as the completion indication. The scenario message describes the game outcome, not a separate measurement of a real device. Preserve the selection while working through this chapter.

Chapter map

Section	Page
Control interpretation	87
Symptom assessment	88
Restoration: steps 1-5	89
Restoration: steps 6-10	90
Acceptance checks	91
Unresolved-fault review	92

Configuration characteristics

This mode requires 5 switches closed and 1 open; 2 socket inputs connected and 2 open. Counts do not identify controls. Matching a count with the wrong controls will not satisfy this mode.

Starting the chapter

Confirm the title, observe the initial state, and read the control interpretation. Treat positions left by a previous participant as observations rather than a known reset.

Mode reference: industrial-robot-arm

Control interpretation

The table maps physical labels to meanings for Industrial Robot Arm. It is a dictionary, not a restoration order or target-state list.

Control	Meaning in this mode
SW1	Safety switch 1 - main power
SW2	Safety switch 2 - E-stop reset
SW3	Toggle switch 1 - servo enable
SW4	Toggle switch 2 - guard door
SW5	Keyed switch 1 - cell access
SW6	Keyed switch 2 - teach pendant key
J1	Socket J1 connected
J2	Socket J2 connected
J3	Socket J3 connected
J4	Socket J4 connected

Use the identifier first

Locate the SW or J label before interpreting the function wording. Two switches of the same type are different inputs. A keyed switch with a specific role here may have a different role in the next mode.

Do not infer a target from a name

A function name does not establish ON or OFF; the procedure supplies the target. A socket description ending in "connected" names the input, which can still be required open.

Symptom assessment: page 88. Restoration starts on page 89.

Symptom assessment

Confirm **Industrial Robot Arm** is selected. Compare observations with the panel. These routes guide inspection; a light does not identify a unique failed component.

Observed condition	Interpretation and next check
Green absent	This mode is not reporting restoration. Inspect all ten conditions.
Blue lit, green absent	Readiness does not establish the other input states. Continue the remaining checks.
Orange lit, green absent	At least one socket is connected. This mode requires 2 connected and 2 open; identity matters.
Adding a socket did not restore the mode	Inspect required-open sockets. More connections are not necessarily progress.
Positions worked in another mode	Re-read this chapter. Targets and functions vary by mode.

Check the assumption

In this mode, 3 of ten inputs must be open. Turning every switch ON and filling every socket cannot produce the required configuration. Use each explicit target.

Choose the next action

If the configuration has not been checked completely, begin on page 89. If every step is complete, use acceptance checks on page 91. Record uncertain readings rather than treating them as correct.

Restoration procedure: 1-5

Apply these steps to **Industrial Robot Arm**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 1 / J4 / Unplug

Locate J4, described as socket j4 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 2 / SW1 / Set ON

Locate SW1, identified as safety switch 1 - main power. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 3 / SW6 / Set ON

Locate SW6, identified as keyed switch 2 - teach pendant key. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 4 / J2 / Plug in

Locate J2, described as socket j2 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 5 / SW3 / Set ON

Locate SW3, identified as toggle switch 1 - servo enable. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Continue on page 90.

Restoration procedure: 6-10

Apply these steps to **Industrial Robot Arm**. Confirm each named control and target before proceeding. Keep earlier corrected controls in position.

Step 6 / J1 / Unplug

Locate J1, described as socket j1 connected. Unplug so this input is **open**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 7 / SW5 / Set ON

Locate SW5, identified as keyed switch 1 - cell access. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 8 / SW2 / Set ON

Locate SW2, identified as safety switch 2 - e-stop reset. Set ON and confirm **ON / closed**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Step 9 / J3 / Plug in

Locate J3, described as socket j3 connected. Plug in so this input is **connected**. Confirm the named socket rather than relying on orange, which indicates only that at least one socket is connected. If this adjustment affects another connection, recheck that input against its own step.

Step 10 / SW4 / Set OFF

Locate SW4, identified as toggle switch 2 - guard door. Set OFF and confirm **OFF / open**. Read the identifier before moving the switch. If it already matches, leave it in place and record the check. This satisfies one input requirement; it does not establish that the whole mode is restored.

Perform acceptance checks on page 91. LED4 green confirms completion.

Acceptance checks

Use this page after all ten restoration steps. A checklist records what you did; the completion signal remains LED4 green.

A. Verify the mode

Read the selection again. It must still be **Industrial Robot Arm**. If it changed, identify the correct chapter before adjusting further.

B. Review the switches

Confirm 5 switches closed and 1 open. Review each SW instruction on pages 89 and 90. Matching the count is insufficient: positions must belong to the named switches.

C. Review the sockets

Confirm 2 inputs connected and 2 open. Inspect J1-J4 against their steps. Orange cannot distinguish the intended combination from an incorrect combination with at least one connection.

D. Observe completion

Check LED4. If green, record the selected mode as restored. If absent, the mode has not passed acceptance even when every written step is ticked off. Continue with unresolved-fault review.

E. Record the outcome

Robot cell restored - supervised automation is ready.

Record the mode, completion indication, and any difference between panel and app. Leave the configuration for inspection. Capture the result before beginning another mode.

Unresolved-fault review: page 92. Restoration record: page 100.

Unresolved-fault review

Use this review when the Industrial Robot Arm procedure appears complete but green is absent, or a previously green indication is lost.

1. Establish what changed

Record whether the mode changed, a switch moved, or a socket was disturbed. Return to the current selection rather than assuming the last successful chapter still applies.

2. Inspect open-state requirements

The open-state inputs are **SW4, J4, J1**. Verify their actions on pages 89 and 90. A connection or ON position at one of these inputs can be the remaining mismatch.

3. Inspect closed-state requirements

The closed/connected inputs are **SW1, SW6, SW3, SW5, SW2, J2, J3**. Verify each against its step. Do not infer condition from one light or the number of cables.

4. Recheck identity

Compare the identifier operated with the instruction. Pay attention to similar switches and adjacent sockets. If a label or position cannot be read, mark it unknown and ask the facilitator to identify it.

5. Preserve evidence

If all ten targets appear correct and green is absent, record the mode, every input, and LED1-LED4. Report disagreement between a visible position and an app-reported condition. Refer further setup diagnosis to the facilitator.

Capture the condition on page 99. Return to page 91 after checking any new adjustment.

Control index: SW1, SW2, SW3

Find the selected mode in the first column. Each entry gives the step number and page containing its instruction. Read the full instruction before operating the control.

Mode	SW1 step / page	SW2 step / page	SW3 step / page
01. Smart Thermostat	3 / 12	7 / 13	1 / 12
02. Smart Lock & Security Panel	5 / 19	3 / 19	8 / 20
03. Cold-Chain Warehouse Sensor	3 / 26	6 / 27	9 / 27
04. Water Pump Station Controller	7 / 34	2 / 33	10 / 34
05. Tactical Field Radio	5 / 40	7 / 41	3 / 40
06. Smart EV Charger	2 / 47	6 / 48	9 / 48
07. Elevator Control Panel	5 / 54	1 / 54	6 / 55
08. Traffic Signal Controller	3 / 61	6 / 62	9 / 62
09. Hospital Patient Monitor	6 / 69	3 / 68	8 / 69
10. Retail Payment Kiosk	4 / 75	8 / 76	1 / 75
11. Drone Ground Control	7 / 83	5 / 82	10 / 83
12. Industrial Robot Arm	2 / 89	8 / 90	5 / 89

Reading an index entry

An entry such as "3 / 12" means step 3 on page 12. It is not a switch position or target value. The referenced instruction supplies the required state and mode-specific meaning.

Use during a recheck

Use the index when a named control needs inspection after the full procedure. It does not replace mode confirmation or the acceptance check. After adjusting, preserve earlier correct settings and inspect LED4.

Control index: SW4, SW5, SW6

Find the selected mode in the first column. Each entry gives the step number and page containing its instruction. Read the full instruction before operating the control.

Mode	SW4 step / page	SW5 step / page	SW6 step / page
01. Smart Thermostat	10 / 13	5 / 12	8 / 13
02. Smart Lock & Security Panel	6 / 20	2 / 19	9 / 20
03. Cold-Chain Warehouse Sensor	1 / 26	8 / 27	4 / 26
04. Water Pump Station Controller	5 / 33	3 / 33	9 / 34
05. Tactical Field Radio	10 / 41	8 / 41	1 / 40
06. Smart EV Charger	4 / 47	3 / 47	8 / 48
07. Elevator Control Panel	10 / 55	8 / 55	3 / 54
08. Traffic Signal Controller	2 / 61	5 / 61	8 / 62
09. Hospital Patient Monitor	5 / 68	1 / 68	9 / 69
10. Retail Payment Kiosk	6 / 76	10 / 76	2 / 75
11. Drone Ground Control	2 / 82	8 / 83	3 / 82
12. Industrial Robot Arm	10 / 90	7 / 90	3 / 89

Reading an index entry

An entry such as "3 / 12" means step 3 on page 12. It is not a switch position or target value. The referenced instruction supplies the required state and mode-specific meaning.

Use during a recheck

Use the index when a named control needs inspection after the full procedure. It does not replace mode confirmation or the acceptance check. After adjusting, preserve earlier correct settings and inspect LED4.

Control index: J1, J2

Find the selected mode in the first column. Each entry gives the step number and page containing its instruction. Read the full instruction before operating the control.

Mode	J1 step / page	J2 step / page
01. Smart Thermostat	2 / 12	6 / 13
02. Smart Lock & Security Panel	7 / 20	1 / 19
03. Cold-Chain Warehouse Sensor	5 / 26	10 / 27
04. Water Pump Station Controller	4 / 33	6 / 34
05. Tactical Field Radio	4 / 40	2 / 40
06. Smart EV Charger	5 / 47	10 / 48
07. Elevator Control Panel	7 / 55	4 / 54
08. Traffic Signal Controller	1 / 61	7 / 62
09. Hospital Patient Monitor	4 / 68	10 / 69
10. Retail Payment Kiosk	7 / 76	5 / 75
11. Drone Ground Control	4 / 82	1 / 82
12. Industrial Robot Arm	6 / 90	4 / 89

Reading an index entry

An entry such as "3 / 12" means step 3 on page 12. It is not a switch position or target value. The referenced instruction supplies the required state and mode-specific meaning.

Use during a recheck

Use the index when a named control needs inspection after the full procedure. It does not replace mode confirmation or the acceptance check. After adjusting, preserve earlier correct settings and inspect LED4.

Control index: J3, J4

Find the selected mode in the first column. Each entry gives the step number and page containing its instruction. Read the full instruction before operating the control.

Mode	J3 step / page	J4 step / page
01. Smart Thermostat	9 / 13	4 / 12
02. Smart Lock & Security Panel	10 / 20	4 / 19
03. Cold-Chain Warehouse Sensor	2 / 26	7 / 27
04. Water Pump Station Controller	8 / 34	1 / 33
05. Tactical Field Radio	9 / 41	6 / 41
06. Smart EV Charger	1 / 47	7 / 48
07. Elevator Control Panel	9 / 55	2 / 54
08. Traffic Signal Controller	4 / 61	10 / 62
09. Hospital Patient Monitor	2 / 68	7 / 69
10. Retail Payment Kiosk	9 / 76	3 / 75
11. Drone Ground Control	6 / 83	9 / 83
12. Industrial Robot Arm	9 / 90	1 / 89

Reading an index entry

An entry such as "3 / 12" means step 3 on page 12. It is not a switch position or target value. The referenced instruction supplies the required state and mode-specific meaning.

Use during a recheck

Use the index when a named control needs inspection after the full procedure. It does not replace mode confirmation or the acceptance check. After adjusting, preserve earlier correct settings and inspect LED4.

Demo cable build reference

For preparing the demo cable bundle. Cable wiring is independent of the Pi harness and does not change the panel receptacle wiring. This is not a mode selection table.

Cable	Pinout	Contacts covered
Cable 1	Side A pin 1 to Side B pin 3 Side A pin 2 to Side B pin 4	J1 and J4
Cable 2	Side A pin 1 to Side B pin 3 Side A pin 2 to Side B pin 1	J1 and J2
Cable 3	Side A pin 1 to Side B pin 3 Side A pin 2 to Side B pin 2	J1 and J3
Cable 4	Side A pin 1 to Side B pin 1 Side A pin 2 to Side B pin 2	J2 and J3
Cable 5	Side A pin 1 to Side B pin 1 Side A pin 2 to Side B pin 4	J2 and J4
Cable 6	Side A pin 1 to Side B pin 2 Side A pin 2 to Side B pin 4	J3 and J4

Plug types

Cables 1-4 use a 2-pin Side A plug and a 4-pin Side B plug. Cables 5-6 use 4-pin plugs on both ends; leave Side A pins 3 and 4 open. Side A and Side B identify the cable ends.

Contact coverage

The last column records the named contact pair associated with a pinout. It does not establish which socket states a mode requires. Use the active chapter for restoration targets.

Panel harness reference

hardware/raspberry-pi/README.md in the repository contains the wiring and GPIO reference. The cable description does not assign a new Pi GPIO or ground connection.

Terminology

Term	Meaning in this handbook
Acceptance check	Final review of mode, inputs, and LED4.
Closed	ON state of a switch input.
Connected	Required plugged-in state of a socket input.
Control identifier	Physical label such as SW3 or J2.
Fault starting state	Deliberately unrestored initial configuration.
Function label	Scenario meaning assigned to an input.
Mode	One of twelve scenarios selected through the app.
Open	OFF for a switch; unplugged/open for a socket.
Restoration target	Required final condition of a named input.
Restore order	Documented method for checking ten inputs.
Selected configuration	Current inputs evaluated for the active mode.
Success indication	LED4 green for the selected mode.
Unknown	Condition not observed or not determined.

These terms describe the demonstration panel. Similar terminology in real devices does not extend the scope of this handbook.

Diagnostic worksheet

Selected mode: _____

Participant / run: _____ Time: _____

Record ON, OFF, or unknown for switches; connected, open, or unknown for sockets. Record observations before suspected causes.

Input	Observed condition	Step checked / note
SW1		
SW2		
SW3		
SW4		
SW5		
SW6		
J1		
J2		
J3		
J4		

Indicator observations

LED1: _____ LED2: _____ LED3: _____ LED4: _____

Last adjustment and observed effect

_____

_____

Uncertainty or facilitator follow-up

_____

Restoration record

Selected mode: _____

Participant / run: _____

Start time: _____ Finish time: _____

Record the result

Check	Observation
Selected mode confirmed	
Steps 1-5 reviewed	
Steps 6-10 reviewed	
Required-open inputs rechecked	
Required-closed inputs rechecked	
LED4 observed green	
Panel and app agreement / difference	

Sections consulted

_____

_____

Questions or uncertainties

_____

_____

Handoff

Leave the observed configuration for inspection. Tell the facilitator whether restoration was confirmed or remains unresolved. A completed form does not substitute for the completion indication.

Facilitator initials: _____ Result recorded: _____